

Tutorial 12

Ranbir Barman, Sukalyan Deb

1 Two dimensional harmonic oscillator

Consider a two dimensional harmonic oscillator described by the Hamiltonian:

$$\hat{\mathcal{H}} = \frac{\hat{p}_x^2}{2m} + \frac{\hat{p}_y^2}{2m} + \frac{1}{2}m\omega^2 x^2 + \frac{1}{2}m\omega^2 y^2 \quad (1)$$

What is the degeneracy of the first excited state. Using degenerate perturbation theory, determine the splitting induced by the perturbation:

$$\delta\hat{\mathcal{H}} = Cxy \quad (2)$$

where C is appropriate constant.

2 Spin- $\frac{1}{2}$ particle under harmonic potential

Consider a spin- $\frac{1}{2}$ particle of mass m moving under the potential $V = \frac{1}{2}m\omega^2 x^2$. The spin of the particle is polarised along the z-axis. Find the energy correction to the ground state if it is perturbed by the Hamiltonian:

$$\delta\hat{\mathcal{H}} = \lambda\hat{x}^2\hat{\sigma}_x \quad (3)$$

3 Variational Principle

Calculate the ground state energy of attractive dirac delta potential $V(x) = -b\delta(x)$, where $b > 0$. The variational trial wave function is given below,

$$\psi(x) = \begin{cases} A \cos\left(\frac{\pi x}{2a}\right), & -a < x < a, \\ 0, & \text{otherwise.} \end{cases}$$